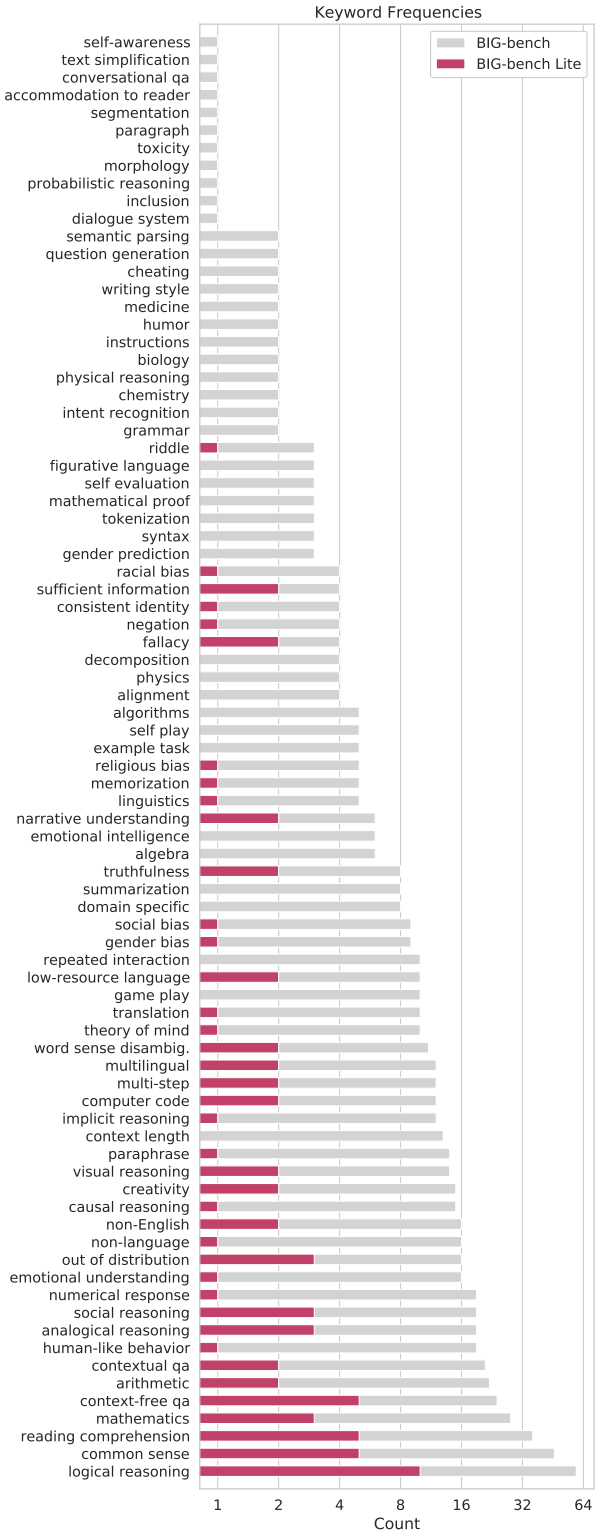

Figure App.5: **Keyword distribution.** Here we show the distribution of keywords for BIG-bench (gray) and for BIG-bench Lite (maroon).

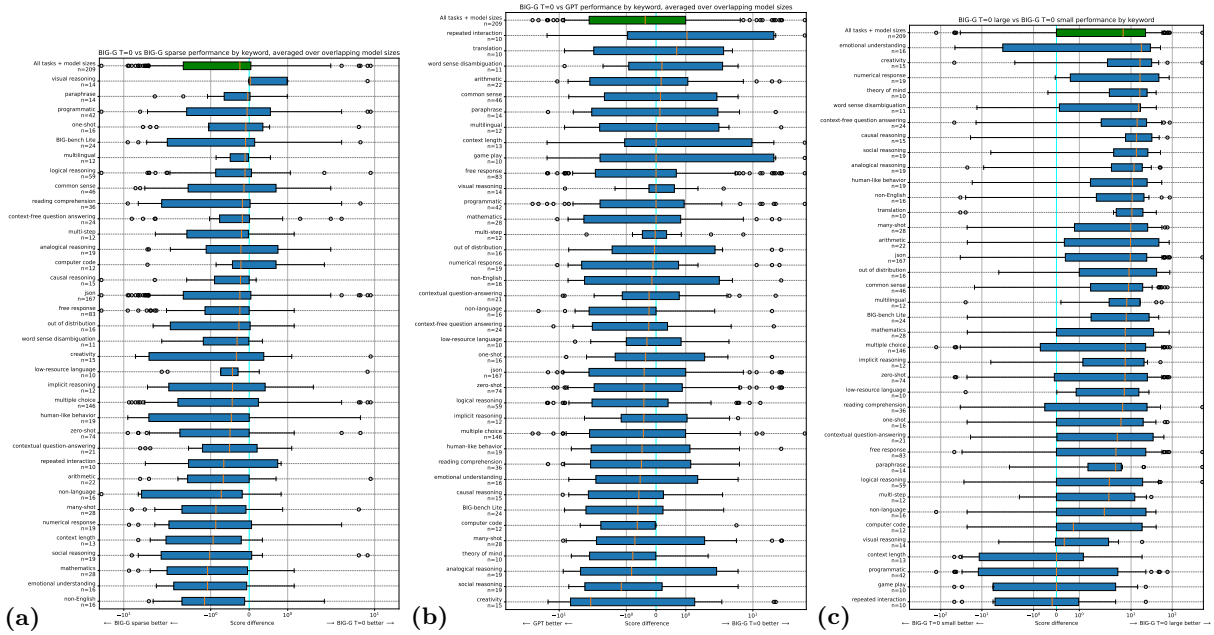

Figure App.6: **Comparing different model classes across keywords.** Box plots show the relative performance of **(a)** BIG-G and BIG-G sparse models, **(b)** BIG-G and GPT models, and **(c)** the largest and smallest BIG-G models. Comparisons are restricted to keywords with at least ten associated tasks, and for **(a)** and **(b)** are also restricted to models of similar scales (specifically, model sizes included cannot exceed the largest or smallest model sizes in the paired model class by more than a factor of two). Performance is averaged over model sizes, and uses the largest shot-count available for all models being compared. Keywords are sorted from largest to smallest median difference in performance, with aggregate performance across all keywords shown in the first row.

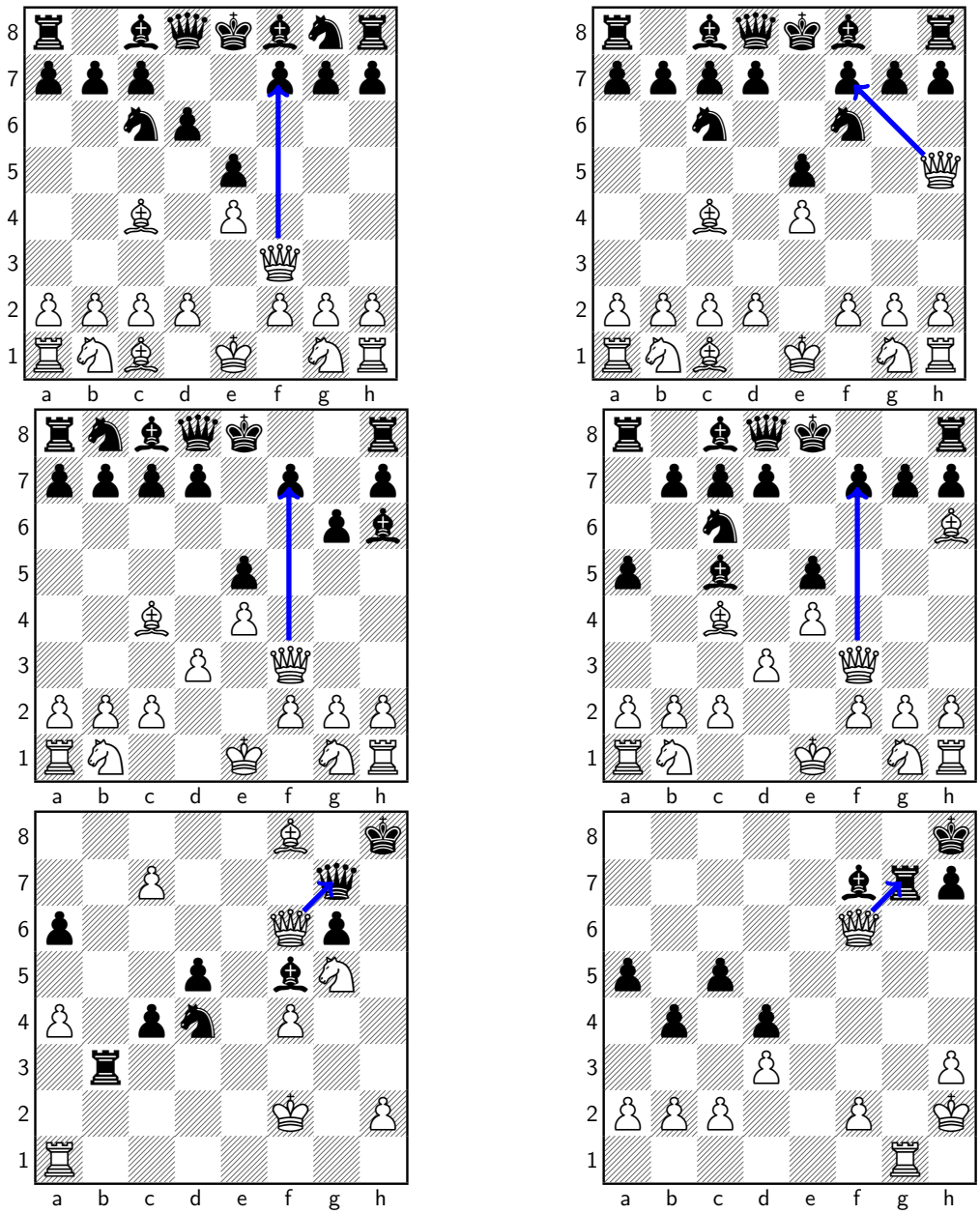

Figure App.7: **The six positions (out of 1,024) in which the largest model found and correctly annotated the checkmating move in a zero-shot setting.** All of them use the queen to perform the mate; the top four are variants of the well-known "scholar's mate."